MAT 504

Homework 1 – From Interatomic Interactions to Effective Material Stiffness

Computer Modelling in Materials Science

Regular deadline: October 12th, 2026 **Late submission:** −20 points per late day

Purpose. In the Week 1 hands-on session, we used the Lennard–Jones potential to connect

$$\text{interaction potential} \longrightarrow \text{force} \longrightarrow \text{microscopic stiffness} \longrightarrow \text{mechanical response}.$$

In this homework, you will extend that analysis to investigate three questions:

1. How do the parameters of an interaction model affect its mechanical response?
2. Over what range is the harmonic/linear approximation valid?
3. How can microscopic stiffness be related to an effective macroscopic material property?

You may reuse functions developed during the hands-on session, but your submitted notebook should be complete and executable independently.

Submission

Submit **one Jupyter notebook** containing:

- all code required to reproduce your results;
- all requested figures and tables;
- concise written answers to the questions;
- appropriate axis labels, legends, and units/reduced units where relevant.

The notebook should run from beginning to end without manual intervention.

Late submissions will receive a **20-point deduction for each late day**.

Assessment of computational work. You are responsible for all code and analysis in your submission. You may be asked in person to explain any part of your notebook, including individual lines of code, equations, numerical procedures, figures, or interpretations. You may also be asked to reproduce or modify a calculation during the discussion. Your ability to explain the submitted work is part of the assessment.

Part 1 – What do the interaction parameters control? 20 points

The Lennard–Jones potential is

$$U(r) = 4\epsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right].$$

Consider the following four models:

Model	ϵ	σ
A	1.0	1.0
B	0.5	1.0
C	2.0	1.0
D	1.0	1.2

1.1 Prediction before calculation – 4 pts

Before running any calculations, predict qualitatively how changing ϵ and σ should affect:

- the equilibrium separation r_0;
- the depth of the potential minimum $U(r_0)$;
- the local pair stiffness k_{pair}.

Write your predictions briefly.

1.2 Numerical comparison – 10 pts

For each model, calculate

$$r_0, \qquad U(r_0), \qquad k_{\text{pair}} = \left.\frac{d^2U}{dr^2}\right|_{r=r_0}.$$

Use the numerical second-derivative procedure developed during the hands-on session for k_{pair}.

Report the results in a single table. Also produce **one figure** showing the four potential-energy curves. Choose the plotting range carefully so that the differences among the models are clear.

1.3 Interpretation – 6 pts

Answer briefly:

a. Which parameter primarily sets the characteristic interaction distance?

b. Which parameter primarily sets the interaction-energy scale?

c. Is a deeper potential-energy minimum necessarily associated with a larger local stiffness? Use your results to justify your answer.

Part 2 – When is the harmonic approximation valid? 25 points

Near its equilibrium separation, the potential can be approximated by

$$U_{\text{harm}}(r) = U(r_0) + \frac{1}{2}k_{\text{pair}}(r - r_0)^2,$$

with corresponding harmonic force

$$F_{\text{harm}}(r) = -k_{\text{pair}}(r - r_0).$$

Use the reference model

$$\epsilon = 1, \qquad \sigma = 1.$$

2.1 Exact and harmonic energy – 7 pts

Consider bond deformations between

$$-10\% \leq \frac{r - r_0}{r_0} \leq 10\%.$$

Plot on the same graph

$$U(r) - U(r_0)$$

and

$$U_{\text{harm}}(r) - U(r_0).$$

Your plot should make the region near equilibrium easy to inspect.

2.2 Exact and harmonic force – 8 pts

Plot on the same graph

$$F_{\text{LJ}}(r)$$

and

$$F_{\text{harm}}(r).$$

Calculate the percentage difference in the **magnitude** of the forces at the following bond strains:

$$-5\%,\ -2\%,\ -1\%,\ +1\%,\ +2\%,\ +5\%.$$

Report the values in a table. For a nonzero force, define

$$\text{relative error} = \frac{\big||F_{\text{harm}}| - |F_{\text{LJ}}|\big|}{|F_{\text{LJ}}|} \times 100\%.$$

2.3 Interpretation – 10 pts

a. Around what deformation range would you regard the harmonic approximation as reasonably accurate for this model? State the criterion you used.

b. Is the deviation from harmonic behavior symmetric between compression and extension?

c. Explain physically why a second-order approximation around r_0 becomes progressively less reliable as the system moves away from equilibrium.

Part 3 – From atomic stiffness to an effective material modulus 35 points

We now return to the one-dimensional chain used in the hands-on exercise. For N atoms with nearest-neighbor interactions,

$$L_0 = (N - 1)r_0.$$

For small deformation,

$$P \approx K_{\text{chain}} \Delta L.$$

The microscopic argument predicts

$$K_{\text{chain}} = \frac{k_{\text{pair}}}{N - 1}.$$

Use the reference interaction

$$\epsilon = \sigma = 1$$

and consider

$$N = 5,\ 10,\ 20,\ 40,\ 80.$$

3.1 Chain stiffness and system size – 12 pts

For each N:

1. construct the chain;
2. calculate its force–extension response over a small strain interval;
3. use
$$|\varepsilon_{\text{strain}}| \leq 0.01$$
to obtain K_{chain} from a linear fit;
4. calculate the prediction
$$K_{\text{chain}}^{\text{pred}} = \frac{k_{\text{pair}}}{N-1}.$$

Prepare a table containing

$$N, \quad L_0, \quad K_{\text{chain}}^{\text{fit}}, \quad K_{\text{chain}}^{\text{pred}}, \quad \text{relative difference.}$$

3.2 Test the scaling law – 8 pts

Plot
$$K_{\text{chain}}$$
against
$$\frac{1}{N-1}.$$

Fit a straight line to your calculated values. Report:

- the fitted slope;
- the fitted intercept.

Compare the fitted slope with k_{pair} and explain what you observe.

3.3 From spring stiffness to an effective modulus – 10 pts

For a continuum rod,
$$P = \frac{EA}{L_0}\Delta L.$$
Therefore,
$$K_{\text{chain}} = \frac{EA}{L_0}.$$

For this toy model, define an **effective modulus**
$$E_{\text{eff}} = \frac{K_{\text{chain}} L_0}{A}.$$

Take
$$A = 1$$
in reduced area units. Calculate E_{eff} for each value of N.

Then show that the microscopic prediction gives
$$E_{\text{eff}} = \frac{k_{\text{pair}} r_0}{A}.$$

Important. E_{eff} here is an effective modulus of our one-dimensional toy model under an assumed cross-sectional area. It should **not** be interpreted as the Young's modulus of a real Lennard–Jones solid.

3.4 Interpretation – 5 pts

Explain why

- K_{chain} decreases as N increases,
- while E_{eff} should be approximately independent of N.

What distinction between a **structural stiffness** and a **material property** does this illustrate?

Part 4 – What can this model actually tell us? 20 points

This final section should be answered concisely. No additional computation is required.

4.1 Model assumptions – 6 pts

Identify **three assumptions or simplifications** in the one-dimensional chain that would have to be reconsidered before using molecular simulation to predict the Young's modulus of a real material.

For each one, write one sentence explaining why it matters.

4.2 Parameters, observables, and assumptions – 6 pts

For each item below, classify it as primarily a **model parameter**, **calculated observable**, or **model assumption**, and briefly justify your choice:

1. ϵ
2. r_0
3. K_{chain}
4. nearest-neighbor interactions only
5. N

If you think an item could reasonably belong to more than one category, explain your reasoning.

4.3 Engineering interpretation – 8 pts

In approximately **150 words**, explain the following statement:

> *A molecular simulation does not simply replace an experiment; it replaces an unknown macroscopic description with a more microscopic model whose own assumptions and parameters must be understood.*

Use the Young's modulus example and at least one result from this homework in your explanation.

Grading

Component	Points
Part 1 – Interaction parameters	20
Part 2 – Harmonic approximation	25
Part 3 – Microscopic stiffness to effective modulus	35
Part 4 – Scientific interpretation	20
Total	**100**

Within each computational section, credit will reflect not only obtaining a numerical result but also:

- correctness of the implementation;
- appropriate figures and labels;
- numerical checks where relevant;
- physical interpretation;
- clarity and reproducibility of the notebook.

A numerically correct result without a defensible analysis may not receive full credit.